

Wenxuan Guo

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Research interests **Experimental Design:** optimal experimental design, covariate balance
Causal Inference: randomization inference, design-based inference
Optimization and Decision-Making: online learning, optimal transport

Education **University of Chicago, Booth School of Business** Chicago, IL
Ph.D. in Econometrics and Statistics, expected 2027 Sep. 2021 – Present
Advisors: Tengyuan Liang and Panos Toulis

University of Chicago, Department of Statistics Chicago, IL
M.S. in Statistics Sep. 2019 – Jul. 2021
Advisor: Lek-Heng Lim

Fudan University, School of Mathematical Sciences Shanghai, China
B.S. in Mathematical Sciences Sep. 2015 – Jul. 2019
Advisor: Shuqin Zhang

Job market paper **Experimental Design When N Equals One** [\[Paper\]](#)
Wenxuan Guo and Tengyuan Liang. 2026.
Under review, Biometrika.

Publications and manuscripts **Gaussianized Design Optimization for Covariate Balance in Randomized Experiments** [\[Paper\]](#)
Wenxuan Guo, Tengyuan Liang, and Panos Toulis. 2026.
Journal of the Royal Statistical Society Series B (JRSS-B).
An earlier version appeared as an extended abstract in the *Proceedings of the ACM Conference on Economics and Computation (EC 2025).*

Reversible Gromov-Monge Sampler for Simulation-Based Inference [\[Paper\]](#)
YoonHaeng Hur, **Wenxuan Guo**, and Tengyuan Liang. 2024.
SIAM Journal on Mathematics of Data Science (SIMODS).

Online Learning to Transport via the Minimal Selection Principle [\[Paper\]](#)
Wenxuan Guo, YoonHaeng Hur, Tengyuan Liang, and Christopher Thomas Ryan. 2022.
Conference on Learning Theory (COLT).

Permutation Inference under Multi-way Clustering and Missing Data [[Paper](#)]

Wenxuan Guo, Panos Toulis, and Yuhao Wang. 2026.
Submitted.

ML-assisted Randomization Tests for Detecting Treatment Effects in A/B Experiments [[Paper](#)]

Wenxuan Guo, JungHo Lee, and Panos Toulis. 2025+.
Submitted.

Invariance-based Inference in High-Dimensional Regression with Finite-Sample Guarantees [[Paper](#)]

Wenxuan Guo and Panos Toulis. 2024+.
Reject & Resubmit, Journal of the American Statistical Association (JASA).

Selected research
in progress

A Coupling Approach to Cost-Aware Online Portfolio Rebalancing
Wenxuan Guo and Tengyuan Liang.

Statistics of the Shepp p -Product
Wenxuan Guo and Lek-Heng Lim.

Awards and
honors

Silver Reviewer Award, International Conference on Machine Learning (ICML)	2026
Master's Program Scholarship, Department of Statistics, University of Chicago	2019 – 2021
Meritorious Winner, Mathematical Contest in Modeling	2019
Selected Member, Top-Notch Student Training Program, School of Mathematical Sciences, Fudan University	2018
Zengshou Xu Scholarship, Fudan University (top 10%)	2018
First Place, Contemporary Undergraduate Mathematical Contest in Modeling	2017
Freshman Scholarship, Fudan University (top 5%)	2015

Software

R package “mwperm” (under development)
Provides finite-sample valid permutation inference for linear models under multi-way clustering, including dyadic and panel-data settings.
GitHub “[gdo_paper](#)”
Provides an R implementation of Gaussianized Design Optimization (GDO), a novel method for achieving optimal covariate balance in randomized experiments.

Teaching experience	Booth School of Business, University of Chicago	
	BUSN 41903: Applied Econometrics (Ph.D. core)	
	– Assistant Instructor	2024 – 2026
	BUSN 41000: Business Statistics (MBA core)	
	– Assistant Instructor	2024 – 2026
	BUSN 41910: Time-series Analysis for Forecasting and Model Building (Ph.D. core)	
	– Teaching Assistant	2025
	Department of Statistics, University of Chicago	
	STAT 34800: Modern Applied Optimization (graduate-level)	
	– Teaching Assistant	2023
Talks	STAT 32940: Multivariate Data Analysis via Matrix Decompositions (graduate-level)	
	– Teaching Assistant	2020
	Experimental Design When N Equals One	
	Econometrics & Statistics Ph.D. Seminar, Chicago, IL	Oct. 2026
	Carnegie Mellon University (CMU) Causal Group (Online)	Sep. 2026
	Permutation Inference under Multi-way Clustering and Missing Data	
	Econometrics & Statistics Ph.D. Seminar, Chicago, IL	Mar. 2026
	Chicago Chinese Social Research Group (CCSRG), Chicago, IL	Feb. 2026
	Gaussianized Design Optimization for Covariate Balance in Randomized Experiments	
	2026 INFORMS Annual Meeting, San Francisco, CA	Nov. 2026
	<i>Session Title: Experimentation</i>	
	Joint Conference on Statistics and Data Science in China (JCSDS 2026), Guizhou, China	Jul. 2026
	American Causal Inference Conference (ACIC 2026), Salt Lake City, UT	May 2026
	Advances with Field Experiments Conference (AFE 2025), Chicago, IL	Sep. 2025
	Joint Statistical Meetings (JSM 2025), Nashville, TN	Aug. 2025
	ACM Conference on Economics and Computation (EC 2025), Palo Alto, CA	Jul. 2025
	Econometrics & Statistics Ph.D. Seminar, Chicago, IL	Apr. 2025
	ML-assisted Randomization Tests for Complex Treatment Effects in A/B Experiments	
	American Causal Inference Conference (ACIC 2025), Detroit, MI	May 2025
	Online Causal Inference Seminar (OCIS)	Dec. 2024
	<i>Discussant: Professor Xinran Li (University of Chicago)</i>	

**Invariance-based Inference in High-Dimensional Regression with
Finite-Sample Guarantees**

Joint Conference on Statistics and Data Science in China (JCSDS), Yunnan, China Jul. 2024

Econometrics & Statistics Ph.D. Seminar, Chicago, IL Jan. 2024

Academic service Reviewer for NeurIPS, ICML, AAAI, *Journal of the Royal Statistical Society Series B* (JRSS-B), *Statistica Sinica*, and *Scandinavian Journal of Statistics*.

Initiated and organized the ChicagoBooth causal inference reading group (2025–2026)

Industry experience	Google <i>Data Science Research Intern on Experimentation</i> • Developed and integrated efficient rerandomization experiments to improve covariate balance. • Implemented bias-minimizing experimental designs for more reliable large-scale hyperparameter tuning. • This experience motivated the subsequent paper, “ <i>Experimental Design When N Equals One</i> ,” which studies optimal design for time-series experiments.	San Bruno, CA May – Aug. 2025
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